1 I

1.1 (1)

Introducing a test function $\psi(x)$, we have:

$$\begin{aligned}[\hat{x},\hat{p}]\psi(x) &= \hat{x}\left(-i\hbar\frac{d\psi}{dx}\right) - \hat{p}(x\psi(x)) \\ &= -i\hbar x\frac{d\psi}{dx} + i\hbar\frac{d}{dx}(x\psi(x)) \\ &= i\hbar\psi(x).\end{aligned}$$

Since this holds for an arbitrary test function, we obtain:

$$[\hat{x},\hat{p}] = i\hbar.$$

1.2 (2)

$$\begin{aligned}a|0\rangle &= 0 \\ \langle x|a|0\rangle &= 0 \\ a\phi_0(x) &= 0 \\ \left(x + \frac{\hbar}{m\omega}\frac{d}{dx}\right)\phi_0(x) &= 0.\end{aligned}$$

1.3 (3)

$$\phi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}.$$

1.4 (4)

$$\epsilon_0 = \frac{\hbar\omega}{2}.$$

1.5 (5)

$$\langle x^2\rangle_0 = \frac{\hbar}{2m\omega}.$$

1.6 (6)

$$\mathcal{P}\phi_0(x) = P_0\phi_0(x).$$

On the other hand, by the definition of the parity operator:

$$\mathcal{P}\phi_0(x) = \phi_0(-x).$$

Since $\phi_0(x) = \phi_0(-x)$, we obtain:

$$\begin{aligned}P_0\phi_0(x) &= \phi_0(x), \\ P_0 &= 1.\end{aligned}$$

1.7 (7)

Assuming that:

$$\mathcal{P}\phi_j(x) = P_j\phi_j(x),$$

and from the definition of the parity operator:

$$\mathcal{P}\phi_j(x) = \phi_j(-x),$$

we have:

$$\phi_j(-x) = P_j\phi_j(x).$$

Replacing x with $-x$, we obtain:

$$\phi_j(x) = P_j\phi_j(-x) = P_j^2\phi_j(x).$$

Thus:

$$P_j = \pm 1.$$

In the operator representation, we have:

$$\mathcal{P}|j+1\rangle = P_{j+1}|j+1\rangle.$$

Since

$$a^\dagger|j\rangle = \sqrt{j+1}|j+1\rangle,$$

we obtain:

$$\begin{aligned}\mathcal{P}a^\dagger|j\rangle &= P_{j+1}a^\dagger|j\rangle,\\ [\mathcal{P}, a^\dagger]|j\rangle &= (P_{j+1} - P_j)a^\dagger|j\rangle.\end{aligned}$$

Under parity transformation, both position and momentum operators change their signs:

$$\begin{aligned}\mathcal{P}\hat{x}\mathcal{P}^{-1} &= -\hat{x},\\ \mathcal{P}\hat{p}\mathcal{P}^{-1} &= -\hat{p}.\end{aligned}$$

Therefore:

$$\begin{aligned}\mathcal{P}a^\dagger\mathcal{P}^{-1} &= \mathcal{P}\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)\mathcal{P}^{-1}\\ &= -a^\dagger.\end{aligned}$$

Hence:

$$\mathcal{P}a^\dagger = -a^\dagger\mathcal{P},$$

and:

$$\begin{aligned}[\mathcal{P}, a^\dagger]|j\rangle &= (\mathcal{P}a^\dagger - a^\dagger\mathcal{P})|j\rangle\\ &= -2P_j a^\dagger|j\rangle.\end{aligned}$$

Comparing the two expressions, we obtain:

$$P_{j+1} = -P_j.$$

Since $P_0 = 1$, we finally get:

$$\boxed{P_n = (-1)^n}.$$

2 II

2.1 (8)

Using the canonical commutation relation, we have:

$$\begin{aligned}\hat{p}e^{ik\hat{x}}|p\rangle &= ([\hat{p}, e^{ik\hat{x}}] + e^{ik\hat{x}}\hat{p})|p\rangle \\ &= (\hbar k + p)e^{ik\hat{x}}|p\rangle.\end{aligned}$$

Thus, $e^{ik\hat{x}}|p\rangle$ is an eigenstate of the momentum operator with eigenvalue $p + \hbar k$. According to the problem description, we identify:

$$e^{ik\hat{x}}|p\rangle = |p + \hbar k\rangle.$$

Since the momentum eigenstates form a complete basis:

$$|n\rangle = \int_{-\infty}^{+\infty} dp\, |p\rangle\langle p|n\rangle,$$

we can write:

$$\begin{aligned}|\psi\rangle &= \hat{U}|n\rangle \\ &= \int_{-\infty}^{+\infty} dp\, \hat{U}|p\rangle\langle p|n\rangle \\ &= \int_{-\infty}^{+\infty} dp\, |p + \hbar k\rangle\langle p|n\rangle \\ &= e^{ik\hat{x}}|n\rangle.\end{aligned}$$

2.2 (9)

The transition probability is:

$$\begin{aligned}P(n \to n) &= |\langle n|\psi\rangle|^2 \\ &= \langle n|e^{-ik\hat{x}}|n\rangle\langle n|e^{ik\hat{x}}|n\rangle.\end{aligned}$$

Under the long-wavelength approximation, we expand the exponential operators up to the second order in k:

$$\begin{aligned}P(n \to n) &\simeq \left(1 - ik\langle x\rangle_n - \frac{k^2}{2}\langle x^2\rangle_n\right) \\ &\quad \times \left(1 + ik\langle x\rangle_n - \frac{k^2}{2}\langle x^2\rangle_n\right) \\ &= 1 + k^2\left(\langle x\rangle_n^2 - \langle x^2\rangle_n\right).\end{aligned}$$

Since the harmonic oscillator eigenstates are also eigenstates of the parity operator, we have:

$$\begin{aligned}\langle x\rangle_n &= \langle n|\mathcal{P}^{-1}(\mathcal{P}\hat{x}\mathcal{P}^{-1})\mathcal{P}|n\rangle \\ &= -\langle x\rangle_n.\end{aligned}$$

Therefore:

$$\langle x\rangle_n = 0.$$

Finally, we obtain:

$$\boxed{P(n \to n) = 1 - k^2\langle x^2\rangle_n}.$$

2.3 (10)

Using the ground state wavefunction obtained in (3):

$$\begin{aligned}\langle 0|e^{ik\hat{x}}|0\rangle &= \int_{-\infty}^{+\infty} \phi_0^*(x) e^{ikx} \phi_0(x)\, dx \\ &= \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{+\infty} e^{-\frac{m\omega}{\hbar}x^2 + ikx}\, dx.\end{aligned}$$

Completing the square and using the Gaussian integral, we obtain:

$$\langle 0|e^{ik\hat{x}}|0\rangle = e^{-\frac{\hbar k^2}{4m\omega}}.$$

Thus:

$$\begin{aligned}P(0 \to 0) &= |\langle 0|e^{ik\hat{x}}|0\rangle|^2 \\ &= e^{-\frac{\hbar k^2}{2m\omega}} \\ &= \boxed{e^{-k^2\langle x^2\rangle_0}}.\end{aligned}$$